

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

University Examination

Date: 05/05/2018

Day: Saturday

Time: 10:00am to 10:30am

Total Marks: 20

Student's ID No.									
Student's Name: _____									
Student's Sign : _____									
Supervisor's Name : _____									
Supervisor's Sign: _____									

Answer Sheet for Multiple Choice Questions (MCQs), True or false and Match the following
Examination : BPT – May 2018
Semester: [8]
Section : I
Course Code : PT402
Course Name: PHYSIOTHERAPY IN NEUROLOGICAL CONDITIONS

Important Instructions:

- There are two sections in the paper. **SECTION –I** contains 20 Questions and **SECTION – II** contains 3 Questions.
- **SECTION –I** should be answered in the question paper itself & **SECTION – II** should be answered in separate answer book.
- Marks on the right side suggest the total marks of that question.
- Maximum time allotted for **SECTION – I** is 30 minutes Question paper for **SECTION – II** will be given once the **SECTION – I** is completed within the stipulated time and returned. However, Candidates who complete **SECTION – I** earlier than 30 minutes can collect main question paper for **SECTION –II** immediately after handing over the **SECTION-I** question paper with answers.
- **SECTION – I** should be attached with the main answer book at the end of the examination and should be submitted to the supervisor.

SECTION-I

Multiple Choice Questions: (MCQs)

(10 × 1 = 10)

(Encircle the correct answer)

1. The physical therapy intervention for a patient with Parkinson's disease includes therapeutic exercise. The desired response to exercise for this patient is:
 - a. Increased isotonic and isometric strength.
 - b. Improved initiation of movement and reciprocal motion.
 - c. Improved range of motion and joint stability.
 - d. Increased proximal and distal stability.
2. A physical therapy plan of care for a newborn with Erb's palsy would include all of the following, EXCEPT:
 - a. Splinting the shoulder in abduction and external rotation, and forearm supination.
 - b. Gentle ROM following the immobilization to prevent contractures.
 - c. Partial immobilization of limb across upper abdomen for 1-2 weeks.
 - d. Facilitate awareness of the involved upper extremity.

(P.T.O.)

3. Your patient is recovering from a right CVA. During gait analysis you observe lateral trunk bending toward the left during midstance on the left leg. Of the following, what would you do to help correct this deviation?
 - a. Strengthen the hip abductors on the right side.
 - b. Strengthen the hip abductors on the left side.
 - c. Strengthen the hip extensors on the right side.
 - d. Strengthen the hip extensors on the left side.
4. You are treating a patient who has a complete spinal cord injury at the level of C6. As a part of your intervention you decide to instruct the family in exercises to maintain the patient's passive range of motion. What should you instruct the family to do?
 - a. Focus on ranging only those muscles that assist with specific functional needs.
 - b. Maintain the normal ROM of all of the patient's muscles.
 - c. Emphasize flexibility of the low back extensors and hamstrings while maintaining normal ROM in the other areas.
 - d. Emphasize range of motion to certain areas to promote function e.g. shoulders.
5. You are taking a patient history, in which the patient tells you he just recovered from a disease but he can't remember the name. When asked to describe his symptoms he tells you that he originally experienced tingling in the hands and feet approximately 3 months after a bout of the flu and that he then experienced progressive weakness to the point that he required a ventilator to breathe. He is now recovering rapidly and is expected to return to a normal functional level in 3 more months. Which of the following conditions did the patient most likely suffer?
 - a. Multiple sclerosis.
 - b. Guillain-Barré syndrome.
 - c. Tick bite.
 - d. Amyotrophic lateral sclerosis (Lou Gehrig disease).
6. After a nerve injury, regeneration occurs proximally first and then progresses distally at a rate of about 1 mm/day. Following a radial nerve injury in the axilla, which muscle would be the last to recover?
 - a. Long head of the triceps.
 - b. Extensor indicis.
 - c. Supinator.
 - d. Pronator teres.
7. A patient complains of a burning sensation in the anterolateral aspect of the thigh. Dysfunction of which nerve could lead to these symptoms?
 - a. Ilioinguinal.
 - b. Genitofemoral.
 - c. Lateral (femoral) cutaneous nerve of the thigh.
 - d. Obturator.

(P.T.O.)

8. A patient with Brown Sequard's syndrome is referred to physical therapy. This type of spinal cord injury presents with:
 - a. Paralysis, loss of proprioception and vibration on the same side as the lesion.
 - b. Loss of pain and temperature on the opposite side of the lesion.
 - c. Paralysis, loss of pain and temperature on the same side as the lesion.
 - d. Both a and b.

9. All of the following are developmental milestones with their appropriate ages, except:
 - a. Sit independently: 5-6 months.
 - b. Walk: 12 months.
 - c. Belly crawl: 3-4 months.
 - d. Cruise: 11 months.

10. A physical therapist forms passive range of motion to a C7 quadriplegic. The patient's bilateral straight leg raise is measured passively to be 90 degrees. What should the physical therapist conclude about the patient's ability to perform activities of daily living?
 - a. The patient requires a straight leg raise of 110 to 120 degrees in order to perform long sit and activities of daily living.
 - b. The patient is at a functional range to perform a long sit and activities of daily living.
 - c. The patient's range of motion is beyond the expected limit for a long sit and activities of daily living.
 - d. The patient requires a straight leg raise of 150 degrees in order to perform long sit and activities of daily living.

Identify the sentence as True or False:

(5 × 1 = 5)

(Please tick mark the answer)

True False

- | | | |
|--|--------------------------|--------------------------|
| 11. Inability to imitate a learned motor task or behavior is known as ideomotor apraxia | ☐ | ☐ |
| 12. Sensory loss, paralysis, loss of bowel & bladder control are classic features of cauda equine syndrome | ☐ | ☐ |
| 13. Multiple Sclerosis is characterized by scattered demyelinating lesions throughout the CNS and PNS and fluctuating symptom severity | ☐ | ☐ |
| 14. The key muscle to assess the function of the Median Nerve is the adductor pollicis muscle | ☐ | ☐ |
| 15. Excessive lateral neck flexion away from the shoulder can lead to Upper brachial plexus lesion. | ☐ | ☐ |

(P.T.O.)

Match the following

(5 × 1 = 5)

16.	Most common cause of automatic hyper-reflexia in a quadriplegic patient	A. Bladder distension
17.	Non- modifiable risk factor of diabetic neuropathy	B. Pressure sore
18.	Most common investigative finding in all types of CP	C. Urinary tract Infection
19.	Most frequent risk of stroke	D. Genetic factor
20.	A frequent complication in complete Spinal cord Injury	E. Hypertension
		F. Poor Glycemic Control
		G. Periventricular Leucomalacia

(16 -), (17-), (18-), (19-), (20-)

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

8th Semester of B.P.T. Examination

May 2018

PT402 Physiotherapy in Neurological Conditions

Date: 05/05/2018

Time: 10:30am to 1:00pm

Maximum Marks: 50

Instructions:

1. There are two Sections in the paper. **SECTION –I** contains 20 Questions and **SECTION – II** contains 3 Questions.
2. **SECTION –I** should be answered in the question paper itself & **SECTION – II** should be answered in separate answer book.
3. Marks on the right side suggest the total marks of that question.
4. Maximum time allotted for **SECTION – II** is 2 Hr 30 min.
5. Draw the figure where necessary.

SECTION-II

Q. 1. Short answers (5 out of 7) (Answer not exceeding 50 words) (5 × 2 = 10)

- a. Advantages and Disadvantages of Modified Ashworth Scale.
- b. Stages of recovery of Stroke.
- c. Types of tremor seen in Parkinson's disease.
- d. Distinguishing features of Cerebellar Ataxia from Sensory Ataxia.
- e. Classification of Brachial Plexus Injury.
- f. Strategies to prevent shoulder pain in stroke.
- g. Common contractures in Spastic diplegic Cerebral Palsy.

Q. 2. Short Notes (4 out of 5) (Answer not exceeding 150 words) (4 × 5 = 20)

- a. Exercises to reduce rigidity in Parkinson's disease.
- b. Glasgow Coma Scale.
- c. Types of Motor neuron Diseases. Clinical presentations of Amyotrophic lateral sclerosis.
- d. Pathophysiology of Multiple sclerosis.
- e. Bladder Management Strategies in Quada Equina Syndrome.

Q. 3. Essay (2 out of 3) (2 × 10 = 20)

- a. Define stroke. Write the management of a 26 year old teacher who has a left CVA of no known cause before 1 week. His goal is to return to his work as a teacher as soon as possible. What are the prognostic indicators of recovery in this case? 2+6+2
- b. Classify Neuropathies. Write the assessment and management of Guillian Barre Syndrome. 2+4+4
- c. Define vestibular rehabilitation therapy. Describe the various exercises of Vestibular Rehabilitation. 2+8